NARAMREDDY CHARAN KUMAR REDDY

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PROFESSIONAL SUMMARY

3rd-year Computer Science student specializing in AI and Machine Learning, building end-to-end systems that reach production rather than stopping at a notebook metric. Experience spans medical-imaging deep learning, gradient-boosted forecasting across thousands of parallel series, and transformers built from scratch — plus the serving layer around them. Strong bias toward leakage-free validation and metrics that describe real behaviour.

TECHNICAL SKILLS

Machine Learning:	XGBoost, LightGBM, Random Forest, Logistic Regression, Scikit-Learn, Optuna.
Deep Learning & CV:	PyTorch, Transfer Learning (ResNet / DenseNet / EfficientNet), CNNs, pydicom, Transformers (Self-Attention), BPE Tokenization.
Data Intelligence:	Pandas, NumPy, Feature Engineering, KNNImputer, Imbalance Handling, RobustScaler.
Engineering & Deploy:	Python 3 (Advanced OOP), Data Structures, FastAPI, REST APIs, SQL (Schema Design), SQLAlchemy, Docker, Azure Container Apps, Git, Claude Code (agentic development).

ENGINEERING PROJECTS

RSNA Pneumonia Detection | **Chest X-Ray Screening | PyTorch, pydicom, FastAPI, React, Docker, Azure** — NaramCharan/RSNA-Pneumonia-Detection

Live application: rsna-app.salmonmeadow-7644e67e.eastasia.azurecontainerapps.io

- Benchmarked DenseNet-121, EfficientNet-B2 and ResNet-34 under an identical two-stage transfer-learning protocol; the fine-tuned ResNet-34 reached **83% pneumonia recall, 0.79 F1 and 86% accuracy** on 1,836 held-out studies, against a 68% always-negative baseline.
- Diagnosed a RandomCrop(224) transform discarding 95% of every 1024×1024 radiograph while retaining the study-level label — removing it produced a larger gain than every architecture change in the project combined.
- Prevented patient-level leakage by splitting on unique patient IDs rather than rows, evaluating the test set once; deployed as one Docker container on Azure serving a React interface and REST API, at $5.06/month after removing a $25.68/month database.

Walmart Store Weekly Sales Forecasting | **Python, Scikit-Learn, XGBoost, LightGBM, Pandas** — NaramCharan/Walmart-Store-Weekly-Sales-Forecasting

- Engineered a global multi-series forecasting pipeline predicting weekly sales across 3,331 store-department combinations, achieving **95.55% validation R^2** with an optimized LightGBM regressor.
- Built a custom recursive walk-forward inference system from scratch to project rolling and lag features over future validation horizons where actual sales are missing.
- Eliminated temporal data leakage by splitting before feature extraction and computing lag/rolling statistics strictly within each partition.

E-Commerce Customer Churn Prediction | **Python, Scikit-Learn, XGBoost, PyTorch, Optuna** — NaramCharan/ecommerce-customer-churn-prediction

- Tuned tree classifiers with Optuna over 10-fold Stratified Cross-Validation, achieving **98.28% accuracy and 94.74% F1** on the selected XGBoost model, with 91.58% recall on a 30% held-out test set.

Handbuilt Transformer | **Learning project | PyTorch, NLP** — NaramCharan/Handbuilt-Transformer

- Built a mini GPT and a mini GPT tokenizer from scratch in PyTorch to learn how ChatGPT-style models work: a ~0.21M-parameter decoder-only Transformer (causal multi-head attention, 4 blocks) trained on Shakespeare, and a GPT-2-style byte-level BPE tokenizer with regex chunking, checked against **tiktoken**; ongoing — next, training on BPE tokens.

Personal Portfolio — naramcharan.me | **Next.js 16, React 19, TypeScript, Three.js, Tailwind v4, GitHub Actions** — NaramCharan/NaramCharan.github.io

Live site: naramcharan.me

- Shipped a WebGL-hero portfolio with interactive project briefs, statically exported and auto-deployed to GitHub Pages; built by **directing Claude Code** across 75+ commits (architecture and product decisions mine, every change reviewed), lifting Lighthouse from 65 to 87.

EDUCATION

B.Tech in Computer Science Engineering (AI & Machine Learning) — 3rd Year — Graduation: May 2028

GD Goenka University — Gurugram, India — Current CGPA: 8.98 / 10.0

CERTIFICATIONS

- Deep Learning Specialization – DeepLearning.AI (Jul 2026)
- Machine Learning Specialization – DeepLearning.AI, Stanford (May 2026)
- Python for Everybody – Univ. of Michigan (Mar 2025)
- Claude Code: Agentic Coding Assistant – DeepLearning.AI, Anthropic (Jul 2026)
- Databases and SQL for Data Science – IBM, Coursera (Nov 2025)
- Prompt Engineering & GenAI – Google, Vanderbilt (Mar 2025)